REBELLIOUS INK · OWN-BRAND PRODUCTS

Stencil transfer gel: formulation brief

For the formulation lab and the PIF. Brief v0.1, 28 October 2026. Prepared for costing, prototyping and safety assessment.

PRODUCT
Stencil transfer gel, 250 ml squeeze bottle

CATEGORY
Leave-on cosmetic, skin preparation (UK Cosmetics Regulation)

WORKING NAME
Rebellious Ink Stencil Lock

PRIORITY
Stencil must survive a full session of wiping. Hold first, cost second, everything else third.

USE
Applied to intact skin before a thermal stencil is pressed on; tattooing begins 10 to 15 minutes later

TARGET COST
Under £0.50 of bulk per 250 ml at pilot scale

1. Design rationale

Commercial transfer gels are humectant solutions (propylene glycol, glycerin) that wet the thermal stencil ink and hold it, with alcohol for flash-off and a gel thickener. Humectants alone stay re-wettable, so the stencil smears under green soap. This brief adds a film former (PVP, or PVP/VA for water resistance) so that once the alcohol flashes off, the stencil ink is locked in a dry flexible film. Glycerin is kept deliberately low to limit smear. No fragrance, because the skin is about to be broken.

2. Base formula (V1)

Phase	INCI name	Trade example	% w/w	Function	Notes
A	Aqua	Deionised water	46.0	Base	q.s. to 100 after adjustments
A	Hydroxyethylcellulose	Natrosol 250 HHR or equivalent	1.0	Thickener	Disperse in water at 40°C with high shear; hydrate 20 to 30 min until clear
A	PVP	PVP K30 (Luvitec K30 or equivalent)	4.0	Film former: stencil hold	Add slowly to the hydrated gel; trial at 3.0, 4.0 and 5.0
B	Propylene Glycol	USP grade	25.0	Humectant and transfer solvent	Primary active for transfer
B	Glycerin	Vegetable, 99.5%	7.0	Humectant and tack	Keep at or below 8; higher smears under wiping
C	Isopropyl Alcohol	IPA 99%	15.0	Flash-off and grip	Add below 30°C; trial at 12, 15 and 20
D	Phenoxyethanol (and) Ethylhexylglycerin	Euxyl PE 9010 or equivalent	1.0	Preservative	Broad spectrum, pH independent; alcohol content supports preservation
D	CI 19140, CI 16035 (or formulator's choice)	Cosmetic grade water-soluble dyes	q.s.	Colour: amber honey	Optional; omit for a clear gel

Water is q.s. so the total is 100.0. Colour, if used, at the lowest level that reads as amber in the bottle.

3. Method of manufacture, pilot 1 kg

1. Phase A: heat water to 40°C. Sprinkle hydroxyethylcellulose under high shear. Hydrate with intermittent stirring 20 to 30 minutes until clear and lump-free.
2. Add PVP to Phase A slowly under stirring until fully dissolved.
3. Phase B: add propylene glycol, then glycerin. Stir to uniform.
4. Cool to below 30°C. Phase C: add isopropyl alcohol under gentle stirring with the vessel covered to limit loss.
5. Phase D: add preservative and colour. Stir 10 minutes. Check pH, viscosity and appearance against section 5.
6. Fill into 250 ml PET squeeze bottles with flip-top caps. Batch code and PAO on the label.

Note for the lab: at 15% IPA, confirm flash point for storage, transport and labelling classification. The product should be stored away from heat and ignition sources either way.

4. Prototype variants

Variant	Change from base	What it tests
V1 (base)	PVP K30 4.0, IPA 15	Reference
V2	PVP K30 3.0	Lower film load: feel and flaking
V3	PVP K30 5.0	Upper film load: maximum hold without flaking
V4	PVP/VA copolymer 4.0 replacing PVP K30	Water-resistant film for long or sweaty sessions. Expected strongest hold
V5	IPA 20, water reduced to balance	Faster set; check the stencil application window does not drop below 30 seconds
V6	IPA 12, water increased to balance	Longer application window; check drying time at 15 min

5. Specification for the approved prototype

Parameter	Target	Method
Appearance	Clear to slightly hazy amber gel, free of lumps	Visual
pH	5.5 to 7.0	Direct, 20°C
Viscosity	Set from the approved prototype; indicative 4,000 to 8,000 mPa·s	Brookfield, spindle and speed to be fixed by the lab
Density	About 1.00 g/ml	Pycnometer
Odour	Faint alcohol, no added fragrance	Olfactory
Microbiology	TAMC < 100 cfu/g, TYMC < 100 cfu/g, specified organisms absent	ISO 21149 / ISO 16212
Challenge test	Pass, criteria A	ISO 11930
Stability	12 weeks at 40°C and 25°C, 3 freeze-thaw cycles, light exposure; no separation, pH drift under 0.5, viscosity within 20%	Lab SOP
Period after opening	12M	Supported by challenge and stability
Shelf life target	24 months unopened	
Pack	250 ml PET squeeze bottle, flip-top, tamper band	Batch code and PAO on label

6. Performance test (decides which variant is approved)

Step	Detail
1	Same drawing, same thermal stencil paper, same printer settings for every sample. Practice skin, cleaned with IPA and fully dry.
2	Thin even layer of gel, stencil pressed for 15 seconds, peeled. Record time to dry-to-touch.
3	At 15 minutes, begin wiping: folded paper towel wetted with 1:8 green soap solution, firm strokes across the lines, same hand, same pressure.
4	Score line integrity 1 to 5 at 10, 25, 50 and 100 wipes (5 = untouched, 3 = faded but usable, 1 = gone).
5	Repeat each sample with the studio's own glide applied to the skin before stencilling, because glide is the usual cause of lift.
6	Benchmarks run alongside: Hornet Stencil and one other commercial transfer gel.
7	Pass mark: score 4 or better at 100 wipes, with glide. Then a real session on a consenting artist's own skin, after the safety assessment is in place, before any client.

7. Regulatory and PIF checklist

- Responsible Person: iLM (existing). Product category: skin care, leave-on, cosmetic. Not a medical device, no medicinal claims.
- Cosmetic Product Safety Report from the qualified safety assessor, covering the PVP and glycol levels and the alcohol content on intact skin. Stencil gel is applied before the skin is broken; state this use clearly so the assessor does not treat it as a wound product.
- Challenge test ISO 11930, stability per section 5, microbiology on each batch.
- UK SCPN notification before first use on any client, including in the studio.
- Manufacture under ISO 22716 GMP with batch records retained.
- Label: product name, function, 250 ml e, INCI list in descending order, PAO 12M symbol, batch code, RP name and UK address, warnings: external use only, avoid eyes, keep away from children, flammable if classification requires it.
- Draft INCI for the label (base formula): `Aqua, Propylene Glycol, Isopropyl Alcohol, Glycerin, PVP, Hydroxyethylcellulose, Phenoxyethanol, Ethylhexylglycerin, CI 19140, CI 16035`
- Claims to substantiate: "stencil stays put through long sessions" needs the wipe-test data on file. Avoid "medical", "surgical", "sterile", "antibacterial".

8. Directions for the label

Shave and clean the area, then dry it completely. Apply a thin, even layer of gel. Place the stencil, press firmly for 15 seconds, peel. Do not touch for 10 to 15 minutes while it sets. Blot, never rub. Work from the bottom up and keep alcohol wipes away from the stencil. External use only. Avoid contact with eyes; rinse with water if it occurs. Keep out of reach of children.